

Vinay Badhan

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Professional Summary

Hands-on seasoned engineer who builds high-leverage systems compressing months of work into days and flipping unit economics from deeply negative to profitable.

At Medable, I lead a globally distributed team of 9 engineers building AI-powered infrastructure for clinical trials. I redesigned study setup end-to-end, cutting delivery from 6 months to under a week and driving margins from -90% to +40% by rethinking the architecture, AI strategy, and how the team operated.

Before that, 5+ years at HackerRank growing from intern to Lead Engineer: cut cloud IDE infra costs by 60%, scaled throughput 10x, and shipped real-time Spark pipelines processing 60M+ submissions per year with sub-2-minute latency.

Technical Expertise

Category	Skills
AI/ML	MCP, Google ADK, LangGraph, Indexing and Embedding, LLM Integration, Evals, Multi-Agent Systems, RAG, Knowledge Graphs
Backend	Python, JavaScript, TypeScript, Node.js, Ruby, Golang, REST APIs, GraphQL, gRPC
Frontend	ReactJS, Next.js, Bootstrap, HTML, CSS, JavaScript, TypeScript
Databases	PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch, DynamoDB
Cloud & DevOps	GCP, AWS, Docker, Kubernetes
Data Engineering	Spark, PySpark, Apache Hudi, AWS Kinesis

Experience

Staff Engineer / Engineering Manager - Medable, Inc

Jun 2023 – Present

Scaled platform reliability and performance wflows to streamline study building and reduce manual effort for clinical trial teams.

- Transformed clinical trial study setup from a 6-month manual process to under a week through end-to-end automation
- Reversed profit margins from -90% to +40% per study by eliminating manual bottlenecks
- Architected self-serve platform enabling sponsors and CROs to independently build and validate studies
- Deployed AI-powered code generation agents es to auto-generate study extensions, tests,and deployments
- Built and scaled a globally distributed engineering team of 9 across India, Argentina, and Dubai, managing hiring, onboarding, and mentorship across time zones
- Conducted systematic load testing across backend services, identified critical bottlenecks, and drove targeted fixes that improved platform concurrency by 68%
- Optimized database queries and introduced efficient indexing strategies, significantly reducing query latency under high-load conditions
- Re-architected traffic distribution across load and faster request processing at scale
- Built a recurring load testing framework, enabling continuous performance regressiondetection before production releases
- Streamlined the release process from weekly to daily by automating documentation generation, reducing manual overhead and accelerating delivery cadence

Lead Engineer - HackerRank

Jan 2022 – Jun 2023

Improved concurrency and scale of assessment platform while leading next-generation IDE design.

- Reduced workspace costs by 60–65% across frenc environments
- Delivered ~\$1.5M annual savings through stration
- Scaled workspace concurrency 10x while impr
- Pioneered next-gen IDE architecture with deieving sub-second load times
- Engineered multi-cloud infrastructure spanning AWS and GCP based on VMs and containers with custom machine images and background processes for exec and IDE
- Elevated IDE launch-time SLA from 80% to 98% via intelligent provisioning and warm-up strategies
- Boosted IDE-Editor launch SLA from 93% to 99% by developing a high-performance management service in Golang

Senior Software Engineer - HackerRank

Apr 2021 – Dec 2021

Led cloud IDE architecture, collaboration solutions for data science, and scaling cloud assessments platform.

- Reduced cloud infrastructure costs by 55% for Developer Skills Platform through architectural improvements
- Improved cloud engineering assessment capacity by 500% via automated warm-up and scaling strategies
- Established comprehensive monitoring infrass and custom dashboards
- Spearheaded editor architecture and real-time collaboration features for data science assessments

Software Development Engineer II - HackerRank

Jan 2020 – Mar 2021

Designed and implemented data ingestion pipeline for HackerRank Enterprise and community platforms.

- Engineered real-time data pipeline processing 60M+ annual submissions with near-instant insights
- Optimized Spark jobs, reducing computation minutes
- Accelerated IDE load times by 50% through Eclipse Theia CDN integration
- Implemented end-to-end observability with logging, alerting, and monitoring dashboards
- Unified data ingestion across interview scorecards, screening scores, challenges, and hackathons

Software Development Engineer I - HackerRank

Jun 2018 – Dec 2019

Integrated cloud IDE for frontend, backend, and data-science assessments in HackerRank for Work platform.

- Pioneered cloud IDE integration for multi-domain technical assessments (frontend, backend, data science)
- Established unit testing framework and back team
- Designed extensible microservice for managing workspace lifecycles
- Contributed to Eclipse Theia open-source project with production-ready enhancements

Awards

- Won AI agentic hackathon at Miami Offsite, Medable 2026
- Barraiser award 2022 at HackerRank for engineering excellence throughout the year
- Spot award in 2021 and 2022 for exceptional quarterly contribution in engineering department
- Won developer hackathon at HackerRank 2022

Key Projects

Tenet AI - Deterministic Audit Ledger for AI Agents (2026)

Stack: Python, FastAPI, React, PostgreSQL, Docker · [GitHub](#)

- Git + Stripe logs for AI agents — captures why agents make decisions, detects policy drift via replay, tracks human overrides, and generates compliance reports
- Enables organizations to audit AI agent behmpliance and traceability for enterprisedeployments where explainability and policy adherence are critical

CRA Agent platform (2026)

- One platform to monitor and conduct multiple study sites across globe
- Reduced the manual and error prone work on logging into multiple third-part systems and check for non-compliance across sites and participants
- Generate reports and draft mails around any risk identified for site monitors

Autobuild — Clinical Trial Study Artifact Generation (2024 - 2025)

Stack: Python, TypeScript, Node.js, GCP

- Generate deployable study artifacts for clinical trials from a JSON spec (based on CDISC DDF industry standard)
- Reduced study artifact generation from days of manual work to seconds, eliminating error-prone handoffs

VM Provider — Multi-Cloud IDE Infrastructure at HackerRank (2022 - 2023)

Stack: Golang, Python, AWS, GCP, Kubernetes, Docker

- Reduced cost per workspace by 59.7% for frontend/backend and 62.95% for data science workspaces
- Increased workspace concurrency by 10x with improved system reliability
- Led design for next-generation IDE by decoupling backend and frontend; implemented multi-cloud support for AWS and GCP

Planb - Routing Engine in Golang (2023)

Stack: Golang, Redis · [GitHub](#)

- Routing engine for maintaining public routes for workspace and workspace exec agent; load tested for 200k concurrency
- Extended open-source project, fixed bugs in Traefik/Redis interaction at scale (>20k records), and deployed to production with comprehensive test coverage

Developer Skills Platform - Real-time Data Ingestion & Analytics at HackerRank (2019 - 2021)

Stack: Spark, PySpark, Apache Hudi, AWS Kinesis, PostgreSQL

- Built system ingesting data from 60M+ submissions yearly with near real-time insights
- Optimized Spark jobs, reducing computationa minutes
- Reduced cloud infrastructure cost by ~55% for Developer Skills Platform